

BrainBolt - Answer Key

Topic: Data Science (15 Questions)

Q1. Which Python library is mainly used for numerical computing with arrays?

- A. Pandas
- B. NumPy <- correct
- C. Matplotlib
- D. Seaborn

Answer: B

Why: NumPy provides the n-dimensional ndarray and is the foundation for numerical computing in Python.

Q2. Which library is best for creating DataFrames?

- A. NumPy
- B. SciPy
- C. Pandas <- correct
- D. Scikit-learn

Answer: C

Why: Pandas introduces the DataFrame — a 2-D labelled tabular data structure.

Q3. What does the term 'overfitting' mean in machine learning?

- A. Model is too simple
- B. Model fits training data too well and fails on new data <- correct
- C. Model has missing data
- D. Model trains too slowly

Answer: B

Why: Overfitting happens when a model learns noise and details of the training data so well that it performs poorly on unseen data.

Q4. Which algorithm is used for classification problems?

- A. Linear Regression
- B. K-Means
- C. Logistic Regression <- correct
- D. PCA

Answer: C

Why: Logistic Regression is a classification algorithm despite its name — it predicts probabilities for class labels.

Q5. What does CSV stand for?

- A. Comma Separated Values <- correct
- B. Common Storage Variables
- C. Compiled Standard Vector
- D. Computed System Value

Answer: A

Why: CSV = Comma Separated Values — a simple plain-text format for tabular data.

Q6. Which measure is most sensitive to outliers?

- A. Median
- B. Mode
- C. Mean <- correct
- D. IQR

Answer: C

Why: Mean is heavily affected by outliers because every value contributes to the sum.

Q7. What is the primary goal of EDA (Exploratory Data Analysis)?

- A. Train a model
- B. Deploy a model
- C. Understand the data before modelling <- correct
- D. Encrypt the data

Answer: C

Why: EDA helps to summarise the main characteristics of data — patterns, anomalies, distributions — before any modelling.

Q8. Which type of plot is best to show the distribution of a single numeric variable?

- A. Scatter plot
- B. Bar chart
- C. Histogram <- correct
- D. Pie chart

Answer: C

Why: A histogram groups values into bins and is ideal for showing distribution of a continuous variable.

Q9. Which sklearn function splits data into train and test sets?

- A. train_test_split <- correct
- B. split_data
- C. kfold
- D. cross_validate

Answer: A

Why: sklearn.model_selection.train_test_split is the canonical helper to split a dataset.

Q10. Which is an UNSUPERVISED learning algorithm?

- A. Decision Tree
- B. K-Means Clustering <- correct
- C. Random Forest
- D. SVM

Answer: B

Why: K-Means groups data into clusters without labels, making it unsupervised.

Q11. What does 'NaN' mean in a Pandas DataFrame?

- A. Not a Name
- B. Not a Number — missing value <- correct
- C. Negative Number
- D. Null Allowed Notation

Answer: B

Why: NaN stands for 'Not a Number' and is used to represent missing or undefined data.

Q12. Which metric is used to evaluate regression models?

- A. Accuracy
- B. Precision
- C. Mean Squared Error <- correct
- D. F1-Score

Answer: C

Why: MSE (and RMSE) measures the average squared difference between predicted and actual values — used for regression.

Q13. Which library is most commonly used for deep learning?

- A. Pandas
- B. TensorFlow <- correct
- C. Statsmodels
- D. Beautiful Soup

Answer: B

Why: TensorFlow (and PyTorch) are the leading deep-learning frameworks.

Q14. What is feature scaling?

- A. Removing rows
- B. Bringing all features to a similar range <- correct
- C. Adding new features
- D. Creating dummy variables

Answer: B

Why: Feature scaling (e.g. standardisation, normalisation) brings features to a comparable range to help algorithms converge faster.

Q15. Which file format is most efficient for big data storage?

- A. .csv
- B. .txt
- C. .parquet <- correct
- D. .docx

Answer: C

Why: Parquet is a columnar, compressed format optimised for analytical workloads on large datasets.